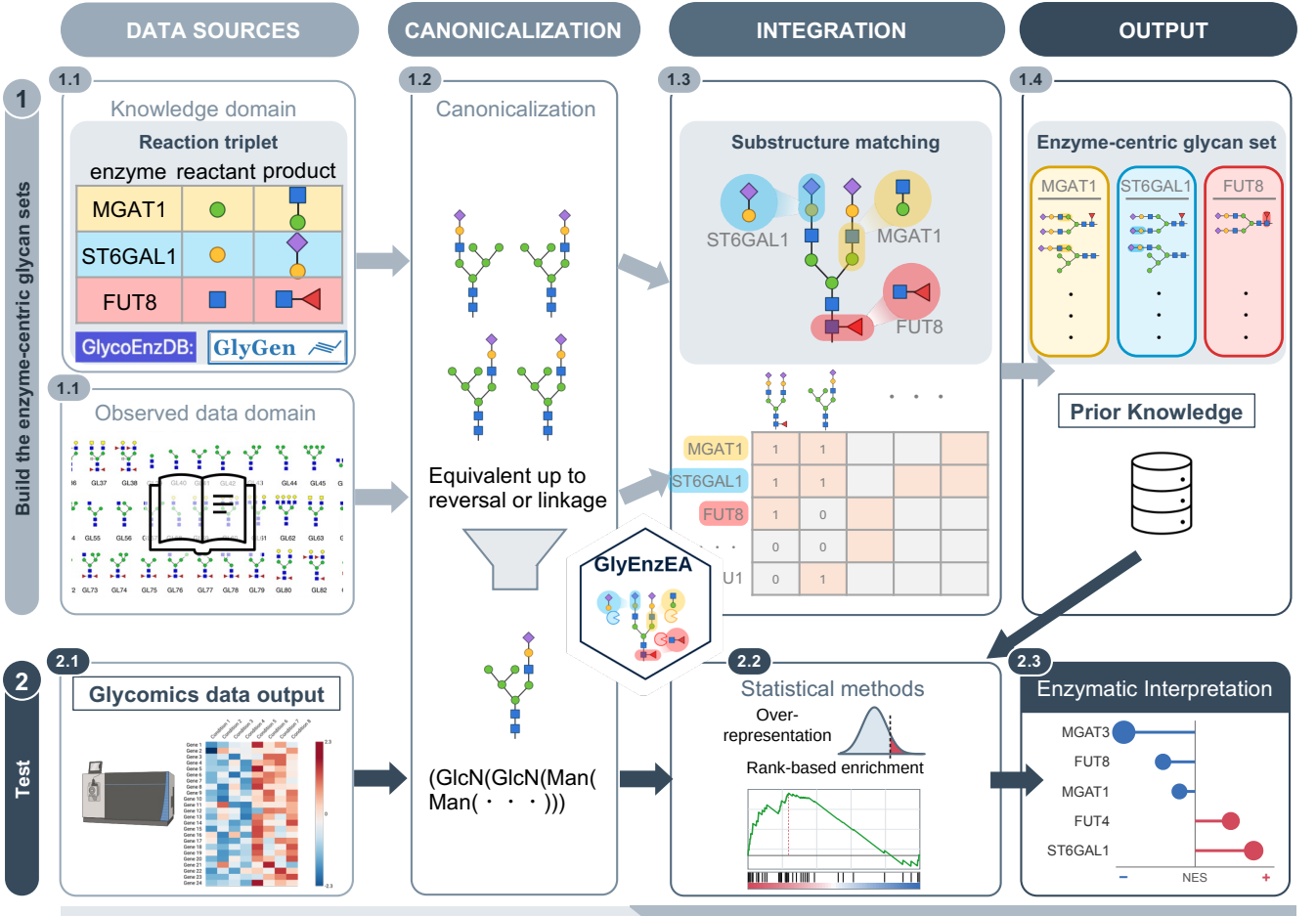

DATA SOURCES
CANONICALIZATION
INTEGRATION
OUTPUT
1
Build the enzyme-centric glycan sets
1.1
Knowledge domain
Reaction triplet
enzyme reactant product
MGAT1
ST6GAL1
FUT8
GlycoEnzDB:
GlyGen
1.1
Observed data domain
1.2
Canonicalization
Equivalent up to reversal or linkage
(GlcN(GlcN(Man(
Man(· · ·)))
GlyEnzEA
1.3
Substructure matching
ST6GAL1
MGAT1
FUT8
MGAT1
ST6GAL1
FUT8
U1
1.4
Enzyme-centric glycan set
MGAT1
ST6GAL1
FUT8
Prior Knowledge
2
Test
2.1
Glycomics data output
2.2
Statistical methods
Over-representation
Rank-based enrichment
2.3
Enzymatic Interpretation
MGAT3
FUT8
MGAT1
FUT4
ST6GAL1
NES
Fragmented knowledge space
Common namespace